NOTE 06 - COMMON PROTOCOLS AND THEIR DEFAULT PORTS

A port is a 16-bit number (0-65535) that identifies which service or process on a host a network conversation belongs to. An IP address finds the machine; the port finds the application on that machine. Ports 1-1023 are the well-known ports and are normally bound by system services.

Main table

Protocol	Port	Purpose
HTTP	80 (TCP)	Transfers web pages and web content in plain text. Requests are readable by anyone on the path.
HTTPS	443 (TCP)	The same as HTTP but wrapped in TLS, so content is encrypted and the server is authenticated with a certificate.
DNS	53 (UDP for normal queries, TCP for large answers and zone transfers)	Resolves domain names into IP addresses and stores other records (mail servers, aliases, verification text).
DHCP	67 (server) and 68 (client), UDP. DHCPv6 uses 546/547.	Automatically gives a host an IP address, subnet mask, default gateway, DNS servers and lease time.
SSH	22 (TCP)	Encrypted remote command-line access, file transfer (`scp`/`sftp`) and tunnelling.
FTP	21 (TCP, control) and 20 (TCP, data in active mode)	Transfers files between a client and server, with no encryption by default.
SMTP	25 (TCP, server-to-server relay). Submission: 587 (with STARTTLS) and 465 (implicit TLS).	Sends and relays email.

Extra ports that appear in the same investigations

Protocol / service	Port	Note
Telnet	23	Plain-text remote login - should not be used; a favourite target for attackers.
TFTP	69 (UDP)	Trivial file transfer, no authentication.
POP3 / IMAP	110 / 143 (unencrypted), 995 / 993 (TLS)	Retrieving mail.
SMB	445	Windows file sharing - common in lateral movement (T1021.002).
RDP	3389	Remote Desktop - common in lateral movement (T1021.001).
Kerberos	88	Domain authentication.
LDAP / LDAPS	389 / 636	Directory services.
MySQL / MSSQL / PostgreSQL	3306 / 1433 / 5432	Database services.
HTTP alternates	8080, 8000, 8888	Common web-server or proxy alternates.

Encrypted versus plain text - why it matters to an analyst

Plain text (readable on the wire)	Encrypted (contents hidden)
HTTP (80), FTP (20/21), Telnet (23), SMTP without TLS (25), DNS (53)	HTTPS (443), SSH (22), SMTPS (465), IMAPS/POP3S (993/995), LDAPS (636)

- With the plain-text protocols a packet capture shows credentials, URLs, file names and email content - which is exactly why attackers still look for them.
- With encrypted protocols the analyst must work from metadata: source and destination IP, port, byte counts, timing, certificate fields and SNI, plus DNS requests that preceded the connection.
- DNS is a special case: it is usually unencrypted, so it is one of the most valuable sources of evidence in a SOC - and one of the most abused (DNS tunnelling, TXT-record C2).

How ports are used during triage

1. Read the destination port first - it usually names the service: `:443` is web traffic, `:445` is Windows file sharing, `:3389` is Remote Desktop.
2. Check whether the port is normal for that host. A workstation making connections to `:445` on twenty other workstations is not normal.
3. Check the direction - inbound connection attempts to `:22` or `:3389` from the internet are worth investigating, outbound `:22` from a server is usually administration or exfiltration.
4. Check for port and protocol mismatches - command-and-control tools often hide on port 443 without doing a real TLS handshake, or run over port 53 with non-DNS content.
5. Remember that ports are defaults, not laws: any service can be moved to any port (`ssh -p 2222`, web servers on 8080), so a port is a strong hint and never proof.

Key points to remember

- HTTP 80, HTTPS 443, DNS 53, DHCP 67/68, SSH 22, FTP 20/21, SMTP 25 (587/465 for submission).
- Ports identify services, not hosts; IP addresses identify hosts.
- Plain-text protocols (HTTP, FTP, Telnet, SMTP, DNS) give an analyst the most evidence and the attacker the most opportunity.
- Unusual ports and protocol mismatches are among the simplest and most reliable detection signals in network traffic.